

Howen VSS Web API

V1.8.5R1

Version Log

Version	Description	Date
V1.0	Initial Document	2020.7.29
V1.1	- Add description of real-time message subscription interface - Added web video integration interface description	2020.9.21
V1.2	- Add device supports specifying feetId; - Add the definition of video request flv http url in Chapter 4; - Add the error code 10043 - Add 2.11 device control web api - Add web player sdk description	2020.11.27
V1.5R1	- Add the description of the location object in the alarm message in the device event message push - Add DMS/ADAS alarm subtype description - Added description of alarm details parameters	2021.2.6
V1.5R2	- Add the description of fast-forward and double-speed parameters to the device video playback URL (supported version: in VSS1.2 and newer) - Add 2.8: Query device GPS and detailed status data by 'page' (supported version: VSS1.5 and newer) - Add 2.10: Query historical alarm details by 'page' (supported version: VSS1.5 and newer) - Chapter 4 add: add the web URL of the server alarm video playback. (supported version: VSS1.2 and newer)	2021.3.5
V1.5R3	- Add: Integration guideline (Chapter 1) - Add: the cookie parameter sent in seconds in the api login interface (3.1) - Add: descriptions of cookie related fields has been added in the interface of 3.8/3.10 - Modify: Equipment alarm event type, upgrade the DMS/ADAS sub-type to the first-level type, the definition of the corresponding first-level type has been added in the equipment event code table (7.9)	2021.3.19
V1.6R1	- Modify: The interface address and request method for querying device GPS and detailed status data by page have changed (the original address and request method are still retained) - Add: more alarm details parameters in "Query historical alarms by page" - Add: interface of obtain device list - Add: interface of download videos in device side - Add: device intercom integration page - Modify: description of location in video file search	2021.4.6
V1.6R2	- Add: the flow chart of Websocket signaling interaction in the [integration guideline chapter] - Modify: Classify the new/modify/delete/device list query interfaces and put them in the device information management interface - Add the interface to obtain the online status of multiple devices - Add the interface for querying the current GPS status of the device - Add the interface for querying alarm evidence - Modify the definition of websocket interface: the login interface does not contain subscription operations, and a separate subscription instruction is added; - Web page embedding method in video integration, support for mobile browser	2021.4.16
V1.7R1	3.11 Video file query interface, add a new function: search center video files 5.5 Server alarm small video playback web page URL, added: playback server recording clip	2021.7.23
V1.8.1R1	- Add: device control command transparent transmission interface (supported by vss1.7), supporting all [device control commands] to be issued to the device, including PTZ, new text distribution and other commands (3.14) - Add: passenger flow statistics report interface (3.16) - Add: driver report interface (3.17)	2021.12.09
V1.8.1R2	3.15 modify the interface request method of web video download	2022.6.22
V1.8.5R1	- Added electronic fence related interfaces - Add more alarm event types	2022.7.19

1 Fundamental Introduction

1.1 API Types

1. http communication interface

The following functions can be achieved through this type of interface:

1. Interface login. Before using all types of APIs, you need to get a license token through this login interface.
2. Obtain historical business data of the VSS platform, such as GPS, alarm events, video records, Video evidence, current state of the vehicle etc.;
3. Synchronize basic operating data, such as synchronizing MDVR device information.
4. Remotely issue control commands to MDVR equipment

2. Websocket communication interface

Through this type of interface, you can subscribe to VSS real-time business data, such as GPS, alarms, online and offline. When VSS receives information such as GPS/alarms reported by MDVR, it can be distributed to subscribers immediately.

3. Javascript SDK

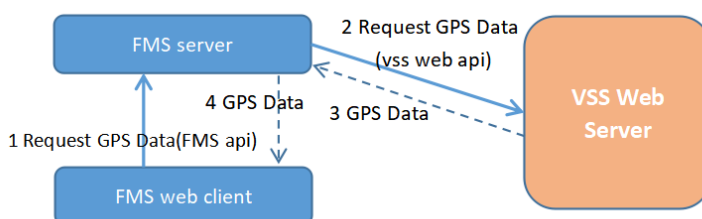
You can integrate the javascript SDK provided by VSS in your web client to, quickly realize video preview and video playback on MDVR and VSS servers.

4. Web embed

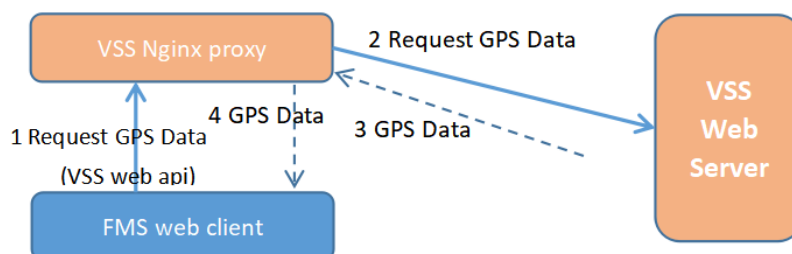
You can directly use the iframe method to embed the VSS web player page in the web page, to quickly realize the video preview and video playback functions on the MDVR and VSS servers.

1.2 Http API Integration Architecture

Architecture 1: Forward data to the FMS client through the FMS backend

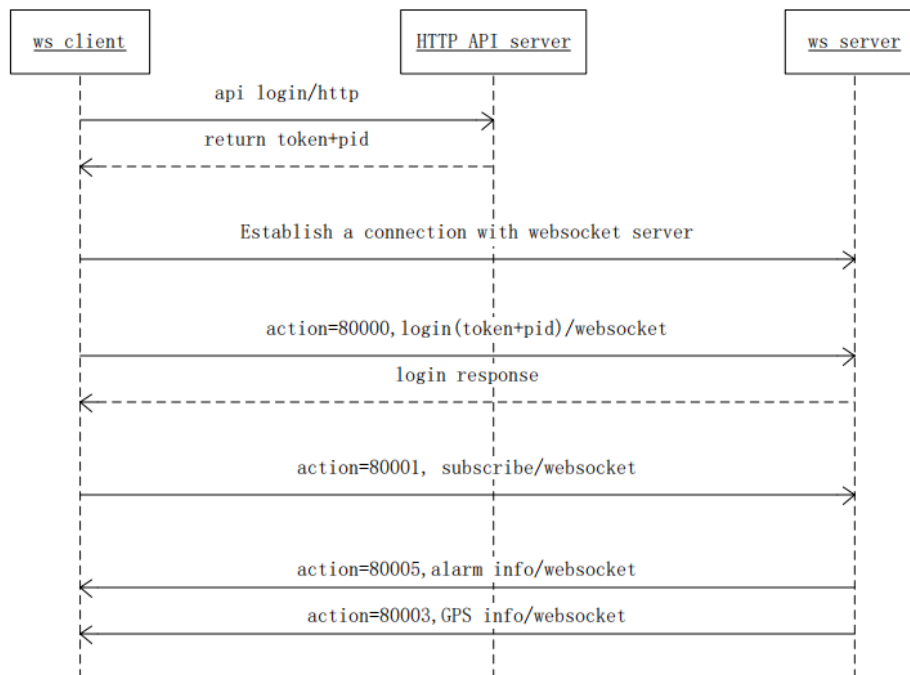


Architecture 2: FMS client directly requests data from VSS (cross-domain)



Note: With this solution, because the browser and VSS Web Server prohibit cross-domain access by default, you need to contact VSS technical support to open the Nginx proxy service. Note: Websocket doesn't have this issue

1.3 Websocket API Signaling interaction process



1. Before the Websocket client connects to the Websocket server, it must obtain the session token and pid parameters through the login interface in the http api.
2. Initiate a connection to the Websocket server and establish a Websocket communication link
3. Client initiates a login instruction to the web server, action=80000
4. Client initiates a message subscription instruction to the web server, action=80001, after completing the message subscription, it waits for the server to actively push GPS/alarm and other messages to the client

2 Transmission Protocol

This interface includes two transmission protocol: HTTP Post and WebSocket.

2.1 HTTP Transmission Protocol

The interface service: HTTP is provided by the VSS platform and follows the HTTP standard protocol (RFC2626)

The default server port: 9966

2.1.1 Get/Post Request Method

1. The Post method support two different data formats.

(1) For sheet submission, shall be "**application/x-www-form-urlencoded**".

Within "HTTP Message Body": para1=value1¶2=value2

(2) JSON

Format in "HTTP Message Body" is as below:

```
{
  "token":"2a545efadfpqoweirsdfs",
  "para1":"value1",
  "para2":"value2"
}
```

Most http interfaces in this document use post commands

2. For Get method, the request parameters are involved in the URL, as below:

[http://192.168.1.2:9966/vss/login.action?
callback= getDataFromVss&token=2a545efadfpqoweirsdfs¶1=value1¶2=value2](http://192.168.1.2:9966/vss/login.action?callback=getDataFromVss&token=2a545efadfpqoweirsdfs¶1=value1¶2=value2)

2.1.2 Common Request Parameters

The request parameters include "Common parameters" and "Private parameters"

Common parameters show as below

Name	Type	Max length	Necessary	Details
token	string	64	yes	Interface access token, all *non-login interface* must carry this parameter, and the value of this parameter is returned after the interface is successfully logged in.
callback	string	32	no	The parameters must be carried, if it is the browser cross-domain access. It will be return the relevant JavaScript execution code, after VSS detects this parameter. (Callback value can only include the letter, number, underline and etc.)

2.1.3 Interface response

JSON type

For the domain access, the JSON data returned in the Message Body. MIME type:

Content-type: application/json; charset=UTF-8

JSON data follow the "RFC 4627"

2.1.4 Common response parameters

Name	Type	Max length	Necessary	Details
status	Int		yes	Interfaces return to interfaces. For details, please refer to Return Code Table
data	Json Object		yes	The returned business data object. For details, it will depend on the definition of each interface.

Common Format as below:

```
{
  "status": 10000, "msg": "Success",
  "data": {
    "para1": "value1",
    "para2": "value2"
  }
}
```

2.1.5 Character Set

Both the request data and the returned data adopts UTF-8

2.1.6 HTTPS

The https server default port is 443 port, and the other request format is the same as http.

2.2 Transfer Protocol: WebSocket

The "WebSocket" service is provided by the VSS platform. ws Port: 36300, wss port :36301 ;

All type of the message subscription and the real-time message pushing interfaces adopt the "WebSocket" protocol (RFC6455). In addition, the transmission data adopts JSON format. (The messages described in this document mainly refer to device GPS message, Alarm message, Online /Offline message and soon)

3 Http Interface Definition

3.1 API Access Login in

Interface address: <http://192.168.1.2:9966/vss/user/apiLogin.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameter s	Type	Length	Necessar y	Details	Example Value
username	string		yes	User name	admin
password	string		yes	Password md5 string	3392e4fcc7d748b04a7046b897d75as

JSON format example:

```
{
  "username": "admin",
  "password": "3392e4fcc7d748b04a7046b897d75as"
}
```

Request succeed, return data example:

```
{
  "status": 10000, // Return status code, refer to return code table
  "msg": "Success", // Indication message
  "data": { //Return data
    "token": "3df52ffe70314f69e7ecad41350eb78e" //token for api request
    "pid":
    "b9c7f86sdeg6f7e4799cefgaaeed465dc1cc704cb1db207bdab9c1758ad39bd187d350fa949b2ff9ddda8436467b1" //pid, used to subscribe to WebSocket request for real-time messages
  }
}
```

Descriptions of Return parameters

Parameter	Type	Length	Necessary	Description
token	string		Yes	The http interface token obtained after successful login. Needs to carry this token in subsequent api interfaces. Token validity period: 30 minutes, if there is no interface interaction within 30 minutes, the token will expire
pid	string		Yes	Websocket interface token

Return cookie parameter description

Parameter	Type	Length	Necessary	Description
JSESSIONID	string	40	Yes	Example value:

D				DB8E82C963EC9EC0E9E92C864FC1B0AB Some interfaces need to use the session mechanism (rely on cookies), and this parameter needs to be carried in the cookie field in the http request header: Cookie: JSESSIONID=DB8E82C963EC9EC0E9E92C864FC1B0AB
---	--	--	--	---

Others return data example:

```
{
  "status": 10003, // Status code, refer to the return code table
  "msg": "Server error", //Indication
  "data": null
}
```

3.2 API Login Out

Interface address: <http://192.168.3.15:9000/vss/user/apiLogout.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
username	string		yes	User name	admin
token	string		yes	Token	

Request succeed return data example:

```
{
  "status": 10000, // Return status code, refer to return code table
  "msg": "Success", //Return Indication
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server Error",
  //Indication
  "data": null
}
```


3.3 Interface of Device information management

Device information management interface, used for real-time synchronization and management of device information between the external platform and the VSS platform.

3.3.1 New Equipment

Interface address: <http://192.168.1.2:9966/vss/vehicle/apiAddDevice.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	yes	Device ID	Fleet001
deviceName	string	64	yes	Device Name	
deviceType	string		yes	Device Type code. For details, please refer to Device Type code	Hero-ME41-04
channelName	String	256	no	Channel Names Set. Channels need to be separate by semicolons (;)	ch1;ch2;ch3;ch4
fleetId	String	255	No	Fleet ID, if it is empty, means it belongs to default fleet	fleet001 Note: The account must obtain this fleet ID authority
token	String		Yes	Token	
ioname	String		No	io name, each io is separated by a semicolon	io1;io2;io3...;io12

Json format example of request:

```
{
  "token": "2sdfgyftwret2asdfhgyh",
  "deviceID": "99990001",
  "deviceName": "90001",
  "deviceType": "Hero-ME41-04",
  "channelName": "ch1;ch2;ch3;ch4",
}
```

Request succeed return data example:

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server error", //Indication
  "data": null
}
```

3.3.2 Modify Equipment

Interface address: <http://192.168.1.2:9966/vss/vehicle/apiModifyDevice.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	details	Example value
deviceID	string	64	yes	Device ID	v001
deviceName	string	64	yes	Device Name	
channelName	String	256	no	Channel Names Set. Channels need to be separate by semicolons(,)	ch1;ch2;ch3;ch4
deviceType	string		yes	Device Type code. For details, please refer to Device Type code	Hero-ME41-04
token	String		yes	Token	

Json format example of request:

```
{
  "token": "2sdfgfytwret2asdfhgyh",
  "deviceID": "99990001",
  "deviceName": "90001",
  "channelName": "ch1;ch2;ch3;ch4",
}
```

Request succeed return data example:

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server error", //Indication
  "data": null
}
```

3.3.3 Delete Equipment

Interface address: <http://192.168.1.2:9966/vss/vehicle/apiRemoveDevice.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
deviceID	string	64	yes	Device ID	Fleet001

Json format example of request:

```
{
  "token": "2sdfgytwret2asdfhgyh",
  "deviceID": "99990001",
}
```

Request succeed return data example:

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server error", //Indication
  "data": null
}
```

3.3.4 Obtain device list by page

interface address: <http://192.168.1.2:9966/vss/vehicle/findAll.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
pageNum	number		Yes	Page number, set to -1 when querying all	2
pageCount	number		Yes	Number per page, set to -1 when querying all	20
isOnline	string		No	Check whether the device is online. 1- means only get online devices 0- means to obtain offline devices. If it is empty, query all devices	1
keyword	string		No	Search for the keyword. Search by device id or device name	
fleetid	String		No	Fleet ID, multiple fleet IDs are separated by commas	

Json format example of request:

```
{
  "token":"2sdfgytwret2asdfhgyh",
  "isOnline":"1",
  "pageCount":"20000",
  "pageNum":"1",
  "keyword":""
}
```

Returned parameter by page:

Parameters	Type	Necessary	Details	Example Value
totalCount	number	Yes	Total number of records	
pageCount	number	Yes	Number of records per page	
fromCount	number	Yes	The starting record number of the current page	
toCount	number	Yes	The last record number of the current page	
totalNum	number	Yes	total pages	
pageNum	number	Yes	The current page number, starting from 1	

Return the attribute parameter table of each device:

Parameters	Type	Length	Necessary	Details	Example Value
deviceno	number		Yes	Device ID, unique	

devicename	number		Yes	Device name	
devicetype	number		No	Device type	
fleetid	String		Yes	Fleet id	
createtime	String		Yes	Device creat time	
videoencodernumber	number		Yes	Number of video channels of the device	
speakernumber	String		Yes	Number of device intercom channels	
accessmode	number		Yes	The online status of the device. Refer: Device status table	
lastonlinetime	String		Yes	Last online time	
lastofflinetime	String		Yes	Last offline time	
longitude	String		Yes	The longitude of the last positioning	
latitude	String		Yes	The latitude of last positioning	
dayUsed	number		Yes	Data cost today, unit: Byte	
monthUsed	number		Yes	Data cost this month, unit: Byte	

The format of the data returned after the request is successful:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "totalCount": 1985,
    "pageCount": 100,
    "fromCount": 1,
    "toCount": 100,
    "totalNum": 20,
    "pageNum": 1,
    "dataList": [
      {
        "guid": "B5A41067-6FB24946-9D4C7E3F-140A115E",
        "deviceno": "00000000",
        "devicename": "00000000",
        "devicetype": "Hero-ME41-02",
        "channelname": "ch1;ch2",
        "sim": "",
        "fleetid": "4A4FE94D-F98A4CDC-B040ED9E-1790838C",
        "remarks": "",
        "createtime": "2021-03-18 11:25",
        "modifytime": "2021-03-18 11:25",
        "videoencodernumber": 2,
        "speakernumber": 1,
        "inputnumber": 2,
        "outputnumber": 2,

```

```

        "accessmode": -1,
        "addressinternet": null,
        "userid": null,
        "username": null,
        "lastonlinetime": "2021-04-01 11:49",
        "lastofflinetime": "2021-04-02 00:47",
        "roleid": null,
        "lastStatusJson": "",
        "longitude": 113.946854,
        "latitude": 22.557869,
        "altitude": 0.0,
        "speed": "0",
        "precision": "0",
        "direct": "0",
        "satellites": "0",
        "mode": "1",
        "dtu": "2021-04-02 00:46:57",
        "todaymileage": 0.00,
        "totalmileage": 0.00,
        "flowLimitDay": -1,
        "flowLimitMonth": -1,
        "flowSettlementDay": 1,
        "dayUsed": 29928,
        "monthUsed": 202239064
    },
    ]
}

```

Examples of returned data in other cases

```

{
  "status": 10003,
  "msg": "Server Error",
  "data": null
}

```

3.4 Get Online Status of single Device

Interface address: <http://192.168.1.2:9966/vss/onoffline/queryDeviceStatus.action>

Request method: [http Post](#)

Data format: [application/json](#) or [application/x-www-form-urlencoded](#)

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
deviceID	string	64	yes	Device ID, multiple IDs separated by commas	device001

Json format example of request:

```
{
  "token": "2sdfgytwret2asdfhgyh",
  "deviceId": "99990001",
}
```

Private response parameters:

Parameters	Type	Length	Necessary	Details	Example Value
deviceId	string	64	yes	Device ID	Fleet001
queryTime	String		yes	Query Time	
accessMode	Int		yes	Device online status code. For details, please refer to Device status code set	
lastOnlineTime	String		no	Last online time	
lastOfflineTime	String		no	Last offline time	

Request succeed return data example:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "queryTime": "2020-07-06 12:12:12",
    "accessMode": -1,
    "lastOnlineTime": "2020-07-06 12:12:12",
    "lastOfflineTime": "2020-07-06 14:12:12"
  }
}
```

Others return data example:

```
{
  "status": 10003, "msg": "Server Error",
  "data": null
}
```

3.5 Get the online status of multiple devices

Interface Address: <http://192.168.1.2:9966/vss/onoffline/queryMoreDeviceStatus.action>

Request Method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Len	Necessary	Description	Example
deviceID	string	64	Yes	Device number, separated by commas	99990001,99990002

json format example of Request:

```
{
  "token": "2sdfgytwret2asdfhgyh",
  "deviceID": "99990001,99990002",
}
```

Private request parameters:

Parameters	Type	Len	Necessary	Description	Example
deviceID	string	64	Yes	Device number	Fleet001
queryTime	String		Yes	Query time	
accessMode	number		Yes	Device online status code Refer: Device status	
lastOnlineTime	String		No	Last online time	
lastOfflineTime	String		No	Last offline time	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "queryTime": "2020-07-06 12:12:12",
      "accessMode": -1,
      "lastOnlineTime": "2020-07-06 12:12:12",
      "lastOfflineTime": "2020-07-06 14:12:12"
    }
  ]
}
```

Return of Other conditions:

```
{
  "status": 10003,
  "msg": "server error",
  "data": null
}
```

3.6 Get the current GPS status of the device

This interface is used to obtain the last GPS and status information reported by the device.

Interface Address: <http://192.168.1.2:9966/vss/vehicle/getDeviceStatus.action>

Request Method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Len s	Necess ary	Description	Example
token	String		Yes	Token returned after successful login	
deviceID	string		Yes	Device number, multiple devices are separated by commas	300330,99990001

json example of Request

```
{
  "token":"f02cfa1cd21138ac89f7f1a32f8339be",
  "deviceID":"4104",
}
```

Private response parameters:

The returned GPS status object parameters are as follows:

Parameters	Type	Lens	Necessary	Description	Example
satellites	number		Yes	Satellite number	
direct	number		Yes	The direction of the vehicle, 0-360 degrees. The true north direction is 0 degrees, increasing clockwise	
speed	number		Yes	Speed, unit km/h	
altitude	number		No	Altitude, unit meter	
longitude	String		Yes	longitude	
latitude	String		Yes	latitude	
precision	number			precision, unit : meter	
createtime	String			Gps time	
accState	number			1-Acc on 0- Acc off	
speed	String			speed	
recordState	number			Channel recording status. Bit0 represents the first channel	
videoMaskState	number			Video occlusion status Bit0 represents the first channel	
videoLostState	number			Video loss status. Bit0 represents the first channel	
ioState	number			io status.	

				Bit0 represents the first channel	
oilVolume	Number			Fuel amount	
tempeAndHumidity	Array Object			Temperature and humidity sensor data array	
isLater	Number			Whether to retransmit data	
reportTime	String			Time to report to the platform	2021-03-02 11:19:38

The data format returned by the request is as follows:

```
{
  "status": 10000,
  "msg": "Success",

  "data": [
    {
      "guid": "D02B567A-3ACD4C80-A1FC05C6-38F8BB29",
      "deviceguid": "10001",
      "deviceName": "10001",
      "longitude": 113.946609,
      "latitude": 22.5581837,
      "altitude": 52.0,
      "createtime": "2021-03-02 11:19:38",
      "speed": "0",
      "precision": "1",
      "direct": "36",
      "satellites": "3",
      "mode": "1",
      "urgency": "0",
      "stateJson": "{\"mileage\":...}", //all State raw data json
      "overSpeed": "0",
      "accState": "1",
      "lowSpeed": "0",
      "recordState": "255",
      "ioState": "1",
      "oilVolume": 0.0,
      "videoMaskState": "0",
      "videoLostState": "252",
      "tempeAndHumidity": [
        "0",
        "0",
        "0",
        "0",
        "0",
        "0"
      ],
      "isLater": 0,
    }
  ]
}
```

```

    "reportTime": "2021-03-02 11:19:39",
  },
]
}

```

Examples of returned data in other cases

```

{
  "status": 10003,
  "msg": "Server Error",
  "data": null
}

```

3.7 Query Device GPS Track by Page

This interface is used to query the historical GPS track of the device

Interface address: <http://192.168.1.2:9966/vss/track/getTrackList.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
pageNum	number		yes	Page number. Set to -1 when querying all	2
pageCount	number		yes	Number per page. Set to -1 when querying all	20
deviceID	string		yes	Device ID	300330
beginTime	string		yes	Begin time	2017-01-23 12:12:12
endTime	string		yes	End time	2017-01-23 13:12:12

Json format example of request:

```

{
  "token": "2sdfgytwret2asdfhgyh",
  "deviceID": "99990001",
  "pageNum": "1",
  "pageCount": "200",
  "beginTime": "2017-01-23 12:12:12",
  "endTime": "2017-01-23 12:12:12"
}

```

Private response parameters:

In the return JSON data object, include all GPS location object and Mileage object array.

Location object parameters show as below:

Parameters	Type	Length	Necessary	Details	Example Value
satellites	number		yes	Number of satellites	
direct	number		yes	Vehicle direction, 0- 360 degrees. The north is 0 degree, increasing clockwise.	
speed	number		yes	Speed. Unit: km/h	
altitude	number		yes	Altitude. Unit: meter	
longitude	String		yes	Longitude	
latitude	String		yes	Latitude	
precision	number			Accuracy. Unit: meter	
dtu	String			GPS time	

Mileage object parameters show as below:

Parameters	Type	Length	Necessary	Details	Example Value
todayDay	number		yes	Total mileage today. Unit: 10 meters	
total	number		yes	Total mileage. Unit: 10 meters	

Request succeed return data example:

```
{
  "status": 10000,          // Return status code, refer to return code table
  "msg": "Success",        //Return indication
  "data":
  [
    {
      "deviceID": "300331",
      "location": {
        "satellites": "0",
        "direct": "-45",
        "speed": "2",
        "altitude": "221",
        "longitude": "114.113586",
        "latitude": "22.5336876",
        "precision": "50",
        "dtu": "2018-03-13 11:30:06",
        "mode": "1"
      },
      "mileage": {
        "todayDay": "100",
        "total": "20000"
      }
    },
  ],
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server Error",
  "data": null
}
```

3.8 Query device GPS and detailed status data by page

This interface is used to query the historical GPS track and corresponding status data of the device

Interface address: <http://192.168.1.2:9966/vss/track/getApiTrackList.action>

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameter	Type	Length	Necessary	Details	Example value
token	String		yes	Token returned after successful login	
pageNum	number		yes	Page number, set to -1 when querying all	2
pageCount	number		yes	Number per page, set to -1 when querying all	20

deviceID	string		yes	Device ID, Multiple devices are separated by commas	300330, 99990001
beginTime	string		yes	Begin Time	2017-01-23 12:12:12
endTime	string		yes	End Time	2017-01-23 13:12:12

Request json example:

```
{
  "token": "f02cfa1cd21138ac89f7f1a32f8339be",
  "deviceID": "4104",
  "beginTime": "2021-03-30 00:00:00",
  "endTime": "2021-04-01 09:00:00",
  "pageNum": "1",
  "pageCount": "200",
}
```

Private response parameters:

In the returned json data object, the parameters are included in the following table:

Parameter by pages:

Parameter	Type	Necessary	Details	Example value
totalCount	number	yes	Total number of records	
pageCount	number	yes	Number of records per page	
fromCount	number	yes	The starting record number of the current page	
toCount	number	yes	The last record number of the current page	
totalNum	number	yes	Total pages	
pageNum	number	yes	The current page number, starting from 1	

The business data is as follows:

Parameter	Type	Length	Necessary	Details	Example value
satellites	number		yes	Number of satellites	
direct	number		yes	The direction of the vehicle, 0-360 degrees. The north direction is 0 degrees, increasing clockwise	
speed	number		yes	Speed in km/h	
altitude	number		No	Altitude, in meters	
longitude	String		yes	longitude	
latitude	String		yes	latitude	
precision	number			Precision in meters	
createtime	String			Gps time	
accState	number			2-Acc on	

				1- Acc off	
speed	String			speed	
recordState	number			Channel recording status. Bit0 represents the first channel, and so on	
videoMaskState	number			Video cover status Bit0 represents the first channel, and so on	
videoLostState	number			Video loss status. Bit0 represents the first channel, and so on	
ioState	number			io status. Bit0 represents the first channel, and so on	
oilVolume	Number			Fuel volume	
tempeAndHumidity	Array object			Temperature and humidity sensor data array	
isLater	Number			Whether to retransmit data or not	
reportTime	String			Time of reporting to the platform	2021-03-02 11:19:38

The format of the data returned after the request is successful is as follows:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "totalCount": 1985,
    "pageCount": 100,
    "fromCount": 1,
    "toCount": 100,
    "totalNum": 20,
    "pageNum": 1,
    "dataList": [
      {
        "guid": "D02B567A-3ACD4C80-A1FC05C6-38F8BB29",
        "deviceguid": "10001",
        "deviceName": "10001",
        "longitude": 113.946609,
        "latitude": 22.5581837,
        "altitude": 52.0,
        "createtime": "2021-03-02 11:19:38",
        "speed": "0",
        "precision": "1",
        "direct": "36",
        "satellites": "3",
        "mode": "1",

```

```

    "urgency": "0",
    "stateJson": "{\"mileage\":\"...\"}", //status raw data
    "overSpeed": "0",
    "accState": "1",
    "lowSpeed": "0",
    "recordState": "255",
    "ioState": "1",
    "oilVolume": 0.0,
    "videoMaskState": "0",
    "videoLostState": "252"
    "tempeAndHumidity": [
        "0",
        "0",
        "0",
        "0",
        "0",
        "0"
    ],
    "isLater": 0,
    "reportTime": "2021-03-02 11:19:39",
},
]
}

```

Examples of returned data in other cases

```

{
  "status": 10003,
  "msg": "server error",
  "data": null
}

```

3.9 Query History Alarm by Page

Interface address: <http://192.168.1.2:9966/vss/alarm/apiFindAllByTime.action>

Request method: [http POST](#)

Data format: [application/json](#) or [application/x-www-form-urlencoded](#)

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
pageNum	number		yes	Page number. Set to -1 when querying all	2
pageCount	number		yes	Number per page. Set to -1 when querying all	20
deviceID	string		yes	Device ID	300330
beginTime	string		yes	Begin time	2017-01-23

					12:12:12
endTime	string		yes	End time	2017-01-23 13:12:12
alarmType	String		no	Alarm event type. Blank mean all type. For details, please refer to Alarm Event Code Table	1

Json format example of request:

```
{
  "token": "2sdfgytwret2asdfhgyh",
  "deviceId": "99990001",
  "pageNum": "1",
  "pageCount": "200",
  "beginTime": "2017-01-23 12:12:12",
  "alarmType": ""
}
```

Private response parameters:

In the return JSON data object, include the Paging parameters and Alarm object array.

Paging parameters show as below:

Parameters	Type	Necessary	Details	Example Value
totalCount	number	yes	Total number of records	
pageCount	number	yes	Number of records per page	
fromCount	number	yes	Starting record number in current page	
toCount	number	yes	End record number in current page	
totalNum	number	yes	Total page number	
pageNum	number	yes	The current page number, starting from 1	

Alarm object parameters show as below:

Parameters	Type	Necessary	Details	Example Value
guid	string		Alarm ID	
deviceId	string		Device ID	
alarmtype	number		Alarm event type. For details, please refer to Alarm Event Code Table	

alarmvalue	string		Alarm details parameters value: Contains detailed alarm related parameters and alarm subtypes. For parameter details, please refer to the code table section;	{"num":"sd1", "st":"0"} 。
createtime	string		Time when Alarm event generated	
alarmState	number		Alarm status.1 mean start, 0 mean end	
AlarmGps	String		GPS coordinates when the alarm occurs	
Speed	String		The speed of the vehicle at the time the warning occurred	113.946564, 22.5582962
Islater	Number		Whether to retransmit data or not	
repoerTime	String		Time for reporting to the platform	2021-03-02 11:19:38

Request succeed return data example:

```
{
  "status": 10000, // Return status code, refer to return code table
  "msg": "Success", //Return
  "data":
  {
    "totalCount": 12, //Total count
    "pageCount":10, //count on each page, -1 mean check all
    "fromCount": 11, //Count starting from current page
    "toCount": 12, // Count ending to current page
    "totalNum": 2, //Total page count
    "pageNum":2, //Current page number
    "dataList":
      [
        {
          "guid": "70814723-2B214A29-98057730-91654774",
            "deviceguid": "333555(333555)",
            "alarmtype": "5",
            "alarmvalue": "io:1; ",
            "createtime": "2021-04-01 09:00:00",
            "alarmGps": "113.946564,22.5582962",
            "speed": "0",
            "takeuptime": null,
            "alarmState": 1,
            "reportTime": "2021-04-01 08:59:55",
            "isLater": 0,
            "takeupUser": null,
            "takeupMemo": null},
        ]
      ]
    }
  }
```

Others return data example:

```
{
```

```

    "status": 10003,
    "msg": "Server Error",
    "data": null
}

```

3.10 Query Device Online/Offline Record by Page

Interface address: <http://192.168.1.2:9966/vss/onoffline/apiFindAll.action>

Request method: [http POST](#)

Data format: [application/json](#) or [application/x-www-form-urlencoded](#)

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example value
pageNum	number		yes	Page number, set to -1 when query all	2
pageCount	number		yes	Number of per page, set to -1 when query all	20
deviceID	string		yes	Device ID	300330
beginTime	string		yes	Begin time	2017-01-23 12:12:12
endTime	string		yes	End time	2017-01-23 13:12:12

JSON format example of request:

```

{
  "token": "2sdfgytwret2asdfhgyh", "deviceID": "99990001",
  "pageNum": "1",
  "pageCount": "200", "beginTime": "2017-01-23  
12:12:12",
  "alarmType": ""
}

```

Private response parameters:

In the return JSON data object, include the paging parameters and online/offline object array.

Paging parameters show as below:

Parameters	Type	Necessary	Details	Example value
totalCount	number	yes	Total number of records	

pageCount	number	yes	Records number per page	
fromCount	number	yes	Starting record number in current page	
toCount	number	yes	End record number in current page	
totalNum	number	yes	Total number of pages	
pageNum	number	yes	The current page number, starting from 1	

Online/offline object array show as below:

Parameters	Type	Necessary	Details	Example value
guid	string		Record	
deviceId	string		Device ID	
createtime	string		Alarm event	
accessmode	number		Online/Offline status. For details, please refer to Device Online Status Table	

Request succeed return data example:

```
{
  "status": 10000, // Return status code, refer to return code table
  "msg": "Success", //Return info
  "data": {
    "totalCount": 12, //Total count
    "pageCount": 10, //count on each page, -1 mean check all
    "fromCount": 11, //Count starting from current page
    "toCount": 12, // Count ending to current page
    "totalNum": 2, //Total page count
    "pageNum": 2, //Current page number
    "dataList": [
      {
        "guid": "0E4732CB-59D74CDA-B410C76D-D3268B65", "deviceId": "999908",
        "accessmode": "-1", "createtime": "2020-08-04 15:49:56"
      },
    ],
  }
}
```

Others return data example:

```
{
  "status": 10003, "msg": "Server
  Error", "data": null
}
```

3.11 Records Searching

Interface address: <http://192.168.1.2:9966/vss/record/videoFileSearch.action>

Request method: [http POST](#)

Data format: [application/json](#) or [application/x-www-form-urlencoded](#)

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	yes	Device ID	Fleet001
startTime	String	19	yes	Starting time	2017-12-20 00:00:00
endTime	String	19	yes	End time	2017-12-20 23:59:59
channelList	String	32	no	The recording channel range, separated by semicolons. The channel number starts from 1.	1;2;3
fileType	Int		yes	1-Ordinary video, 2- Alarm video, 3-Normal picture, 4- Alarm picture	1
location	Int		yes	Video storage location. 1-Device 2-Automatic Download Server 4-Alarm small video server (including http small video and platform alarm triggered small video) 5-Server recording files	1
scheme	string		no	Video download address protocol type. It can only be http or https, default http	https

Example of Request JSON format:

```
{
  "token":"2sdfgytwret2asdfhgyh",
  "deviceID":"99990001", "startTime":"2017-12-20
00:00:00",
  "endTime":"2017-12-2021:00:00",
  "channelList":"1;2",
  "fileType":"1",
  "location":"1"
}
```

Private response parameters:

In the return JSON array object: Files and array element object parameters show as below:

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	yes	Device ID	Fleet001
channel	Int		yes	Channel number start from 1.	
duration	Int		no	Time. Unit: second	
size	Int		no	Files size. Unit: Byte	
type	Int		yes	1-Ordinary video, 2- Alarm video, 3-Normal picture, 4- Alarm picture	
start	String	20	yes	Document starting time	
stop	String	20	yes	Document end time	
path	String	256	no	Files full path	
alarmType	Number		no	Alarm type. refer to: alarm event code table	
downUrl	String	255	no	Video download address. Currently only supports downloading of files such as small videos and pictures of server-side alarms	

Request succeed return data example:

```
{
  "status": 10000,           // Return status code, refer to return code table
  "msg": "Success",         //Return indication
  "data": {
    "files": [
      {
        "deviceID": "300330",
        "channel": "1",
        "duration": "296",
        "size": "97991535",
        "type": "1",
        "start": "2017-12-2000:00:00",
        "stop": "2017-12-2000:04:56",
        "path": "/0000_00870000_5_1513728000_1513728296_0_97991535.h264"
      },
      {
        "deviceID": "300330",
        "channel": "2",
        "duration": "296",
        "size": "9943402",
        "type": "1",
        "start": "2017-12-2000:00:00",

```

```

"stop": "2017-12-2000:04:56",
"path": "/0001_00870200_5_1513728000_1513728296_0_9943402.h264"
"alarmTyp": 132,
  "DownUrl":
    http://47.115.83.226:9988/fileSrv/fileDown.php?filePath=QzovUHJvZ3JhbSBGaWxlcYAo
    eDg2KS9Ib3dlblZzc1NlcnZpY2UsvaHRkb2NzL3Zzc0ZpbGVzL2hmdHAvODg4ODg4M
    S9BSS8yMDIxMDQwMS8xNDsUxNTIfODIvMF82NF84Ml8wXzE2MTcyODg3MTkub
    XA0&token=30fe21f60e9a2d739126665f634976b5
  }
{
  "deviceId": "300330",
  "channel": "2",
  "duration": "296",
  "size": "9943402",
  "type": "1",
  "start": "2017-12-20 00:00:00",
  "stop": "2017-12-20 00:04:56",
  "path": "/0001_00870200_5_1513728000_1513728296_0_9943402.h264"
}
]
}
}

```

Others return data example:

```

{
  "status": 10003,
  "msg": "Server error",
  "data": null
}

```

3.12 Search Alarm Evidence

Interface Address: <http://192.168.1.2:9966/vss/record/evidenceToRetrieve.action>

Request method: [http POST](#)

Data format: [application/json](#) or [application/x-www-form-urlencoded](#)

Private request parameters:

Parameters	Type	Length	Necessary	Details	Example value
conditionName	string	64	Yes	Device keyword, which can be device number or device name	V00001
startTime	String	19	Yes	Query range start time	2017-12-20 00:00:00
endTime	String	19	Yes	Query range end time	2017-12-20 23:59:59
alarmType	String		No	Alarm event type, blank means all types. Refer:	1

				<u>Event of alarm</u>	
scheme	string		No	Request to provide evidence to download the Url protocol type. Could be http or https, default is http	https

Json format example of Request:

```
{
  "token":"2sdfgytwret2asdfhgyh",
  "conditionName":"99990001",
  "startTime":"2017-12-20 00:00:00",
  "endTime":"2017-12-20 21:00:00",
  "alarmType": "",
  "scheme": "http"
}
```

Private response parameters:

Return an array of alarm objects, the parameters of alarm objects are as follows:

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	Yes	device ID	Fleet001
alarmType	Number		Yes	Alarm type. Refer: Alarm of event	
alarmGuid	string	64	Yes	Unique id of alarm message	
alarmGps	string			GPS coordinates when the alarm occurs	113.946564,22.5582962
alarmTime	string		Yes	Alarm occurrence time	2017-01-23 12:12:12
alarmFile	Object array		Yes	The array of evidence files corresponding to the alarm	
alarmFile Object array:					
fileType	number		Yes	1-Ordinary video, 2-Alarm video, 3-Normal picture, 4-Alarm picture	
filePath	String	256	Yes	Path of evidence file	
channel	number			Channel number, starting from 1	
fileThumbnail	string	256		Evidence thumbnail path	
fileStartTime	string		Yes	Starting time	2017-01-23 12:12:12
fileStopTime	string		Yes	End Time	2017-01-23 12:12:12
downUrl	String	1024		Video download address	

Request returned data successfully

```
{
  "status": 10000,
```



```

"msg": "success",
"data": [
  {
    "deviceID": "88888881",
    "alarmGuid": "91BAE324-0B6747A7-BAD0ECED-3EAA4BE0",
    "alarmType": "132",
    "alarmGps": "113.946548,22.5582657",
    "alarmTime": "2021-04-23 13:37:55",
    "alarmFile": [
      {
        "fileGuid": "FB07E391-4FCD44C3-92FEB1BF-475D7188",
        "fileType": "2",
        "filePath": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/88888881/AI/20210423/133755_82/0_64_82_0_1619185075.mp4",
        "downTaskType": "3",
        "channel": 0,
        "fileThumbnail": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/88888881/AI/20210423/133755_82/0_64_82_0_1619185075.jpg",
        "fileStartTime": "2021-04-23 13:37:49",
        "fileStopTime": "2021-04-23 13:38:00",
        "alarmType": "132",
        "downUrl":
"https://47.115.83.1:36301/fileSrv/fileDown.php?filePath=QzovUHJvZ3JhbSBGaWxlcyAoeDg2KS9lb3dlblZzc1NlcnZpY2UvaHRkb2NzL3Zzc0ZpbGVzL2hmdHAvODg4ODg4MS9BSS8yMDIxMDQyMy8xMzM3NTVfODIvMF82NF84Ml8wXzE2MTkxODUwNzUubXA0&token=d648caedf89723d96dc114b0dd43e1ea"
      },
      {
        "fileGuid": "1EB9CA53-B175429E-857BD752-288F3F9D",
        "fileType": "4",
        "filePath": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/88888881/AI/20210423/133755_82/0_64_82_0_1619185075_0.jpg",
        "downTaskType": "3",
        "channel": 0,
        "fileThumbnail": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/88888881/AI/20210423/133755_82/0_64_82_0_1619185075_0.jpg",
        "fileStartTime": "2021-04-23 13:37:55",
        "fileStopTime": "2021-04-23 13:37:55",
        "alarmType": "132",
        "downUrl":
"https://47.115.83.1:36301/fileSrv/fileDown.php?filePath=QzovUHJvZ3JhbSBGaWxlcyAoeDg2KS9lb3dlblZzc1NlcnZpY2UvaHRkb2NzL3Zzc0ZpbGVzL2hmdHAvODg4ODg4MS9BSS8yMDIxMDQyMy8xMzM3NTVfODIvMF82NF84Ml8wXzE2MTkxODUwNzVfMC5qcGc=&token=de7a6331c39317eb3649654b08cb22"
      }
    ]
  },
],
}

```

Other conditions

```

{
  "status": 10003,

```

```

"msg": "Server Error",
"data": null
}

```

3.13 Device control

Interface address: <http://192.168.1.2:9966/vss/vehicle/apiVehicleControl.action>

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters: Json

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	yes	Device ID	dev00001
action	number		yes	1--restart 2--Restore factory settings 3--Pan-tilt 4--Image capture 5--Restore the oil circuit 6--cut off the oil circuit 7--Restore circuit 8--cut off the circuit 9--G-Sensor Calibration	
Detailed parameters of image capture command					
channelList	string	64		Channel list, pass; split, start from 1. Only valid for snapshot	1;3;5;6

Request a json format example:

```

{
  "token": "2sdfgytwret2asdfhgyh",
  "deviceID": "9999999",
  "action": "4",
  "channelList": "1;4;5;6"
}

```

Request returned data successfully

```

{
  "status": 10000,
  "msg": "Success",
  "data": null
}

```

Examples of returned data in other cases

```

{
  "status": 10003,
  "msg": "Server error", //remind information
  "data": null
}

```

3.14 Device control command transparent transmission (supported above vss1.7)

This interface is used to transparently transmit device control commands. The commands are encapsulated in json format, and the vss platform forwards the instructions to the device.

Interface address: http://192.168.1.2:9966/vss/vehicle/apiVehicleCommonControl.action

Request method: http Post

Request data format: json

Private request parameters:

parameter	type	length	Required?	description	Sample value
deviceID	string	64	Yes	device ID	dev00001
action	number		Yes	Command type, currently supports following commands: 1--restart 2--Restore factory settings 3--PTZ 4--Image capture 5--Restore the oil 6--cut off the oil 7--Recovery circuit 8--cut off the circuit 9--G-SENSOR calibration 10- send text	1
Instruction parameter object					
payload	Object			At present, there are only payload parameter objects for commands such as: captured pictures, PTZ, and sent text.	

Description of each command parameter object

Snapshot command parameters		
Content	Field name	Instruction
Channel list	channelList	Pass; split, starting from 1, such as "1;3;5;6" for 4 channels

"Payload":

```
{
  "channelList": "1;4;5;6"
}
```

PTZ control load parameters		
Content	Section name	instruction
Cction	action	refer to: PTZ action code
Channel	channel	0~11 start from 0
X speed	xspeed	Movement speed in X direction, 1~10

Y direction speed	yspeed	Movement speed in Y direction, 1~10
Preset position	preset	0~255

"Payload":

```
{
  "action": "3",
  "xspeed": "5",
  "yspeed": "5",
  "preset": "1",
  "channel": 0
}
```

Send text load parameters		
Content	Field name	Description
type	type	1-Display text 2-Broadcast text
Text content	content	

Example of request json format:

```
{
  "token": "2sdfgytwret2asdfhgyh",
  "deviceID": "9999999",
  "action": "4",
  "payload": { //Command parameter object
    "action": "3",
    "xspeed": "5",
    "yspeed": "5",
    "preset": "1",
    "channel": 0
  }
}
```

Returned data after requesting successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
}
```

```
"data": null
}
```

3.15 Device video download in Web

3.15.1 Add video download task

The client requests the management center to create a web video download task through this interface.

http interface address: http://192.168.1.2:9966/vss/vss/webdownrecord/insert.action

Request method: http post

Data format: Content-Type: application/json

Json format parameters of Request

Parameters	Type	Length	Necessary	Details	Example value
token					The same as above
username	String	32	no	Download user	admin
resStartTime	string	64	yes	The start time of the task start time range. Format: YYYY-MM-DD HH:mm:ss	2021-02-23 12:12:11
resEndTime	string		yes	The deadline of the task start time range	
deviceNo	string		yes	Device number, can be empty	
channel	Number		yes	Channel number, starting from 1	
resDeviceFilename	String			The path of the video file on the downloaded device. If this field has a value, it will be downloaded according to the file, otherwise it will be downloaded according to the time range by default	
fileFormat	String		yes	File format, including: ● mp4 ● Avi Currently only supports mp4	

Json format example of request:

```
{
  "username": "",
  "deviceNo": "girlfriend",
  "channel": "1",
  "resStartTime": "2021-04-06 00:00:00",
  "resEndTime": "2021-04-06 00:02:00",
```

```
"resDeviceFileName": "",
"fileFormat": "mp4",
"token": "f02cfa1cd21138ac89f7f1a32f8339be",
"lang": "zh_CN"
}
```

Request returned data successfully

```
{
  "status": 10000, //return code, refer to Error code for HTTP interface
  "msg": "success", /return message
  "data":
  {
    "taskId": "20198002_20201212102810_20201212103810_1_mp4",
    "fileServerHost": "http://172.16.50.210:9988",
    "resWebPath": "/vssFiles/webdown/20198002/20201212102810_20201212103810_1.mp4",
    "repeat": 0
  }
}
```

Returned data object parameters:

Parameters	Type	Length	Necessar y	Details	Example value
taskId	String	32	yes	The task UUID generated by the server.	
fileServerHost	String	128	yes	Video file server address Format :http://ip:port, or: https	
resWebPath	string	256	yes	The web address of the final video file, which compose a complete video download address with fileServerHost	
repeat	Number		yes	0-represents a new task 1- represents a task that is repeated and has been downloaded 2- Represents the repeated task being downloaded Regardless of whether it is repeated, it should be created successfully, it should be displayed in the task list that is being downloaded, and the download progress should be retrieved.	

Examples of returned data in other cases

```
{
  "status": 10003, //status code, refer to Error code for HTTP interface
  "msg": "Server Error", //remind information
  "data": null
}
```

3.15.2 Video download progress acquisition interface

The client obtains the web video download progress from the management center through this interface.

http interface address: <http://192.168.1.2:9966/vss/webdownrecord/findTaskProgress.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Json format parameters of request

Parameters	Type	Length	Necessar y	Details	Example value
token					
taskId	String	128	yes	Task UUID	
resWebPath	string	256	yes	The web address of the final video file.	

Returned data after requesting successfully

```
{
  "status": 10000, //Return code, refer to Error code for HTTP interface
  "msg": "success", /Return message
  "data":
  {
    "taskId": "20198002_20201212102810_20201212103810_1_mp4",
    "downState": -1,
    "progress": 81,
  }
}
```

Returned data object parameters:

Parameters	Type	Length	Necessar y	Details	Example value
taskId	String	128	yes	Task UUID	
downState	number	128	yes	Download status. 0-not started 1- is downloading 2-download complete -1-Pause state -9-Abnormal stop state When the status is 0/-1/-9, the user can choose to continue downloading	
progress	string	256	yes	The download progress percentage. Integer from 0-	

				100	

Examples of returned data in other cases

```
{
  "status": 10003, // status code, refer to Error code for HTTP interface
  "msg": "Server Error", //remind information
  "data": null
}
```

3.15.3 Video download task management (stop/resume/delete) interfaces

The client uses this interface to manage tasks, including stopping tasks, continuing to download (restore), and delete.

http interface address: <http://192.168.1.2:9966/vss/webdownrecord/taskManagement.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Request json format parameters

Parameters	Type	Length	Necessar y	Details	Example value
token					
username	string	32	yes	Download user	1
taskId	string	128		Task ID	
action	Number		yes	1 means delete task, delete task will delete the corresponding video file. 0 means stop the task 1 means resume download	
resWebPath	string	256	yes	The web address of the final video file.	

Json format example of request:

```
{
  ...
}
```

Request returned data successfully

```
{
  "status": 10000, //Return code, refer to Error code for HTTP interface
  "msg": "success", /returned messages
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003, //Return code, refer to Error code for HTTP interface
  "msg": "Server Error", //remind information
}
```



```
"data": null
}
```

3.15.4 Query video download task list

Interface address: http://192.168.3.15:9000/vss/webdownrecord/findPage.action

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Request parameter

Parameters	Type	Length	Necessary	Details	Example value
pageNum	int		yes	Page number, query all -1	2
pageCount	int		yes	Number per page, query all -1	20
token	varchar		yes	token	
taskState	Int			Task status. 1- Unfinished download 2- The download is complete Empty means search out all	
taskStartTime	string	64	yes	The start time of the task start time range. Format: YYYY-MM-DD HH:mm:ss	2021-02-23 12:12:11
taskEndTime	string			Deadline of the task start time range	
deviceNo	string		yes	Device number, could be empty	
username	String	32	yes	Download user	
scheme	String	32		User browser protocol scheme. It can only be http or https	

Request returned data successfully

```
{
  "status": 10000, //return code, refer to Error code for HTTP interface
  "msg": "success", /return information
  "data": {
    "totalCount": 11, //total number
    "pageCount": -1, //total number per page, -1 means query all=
    //Start number of current page
    "toCount": -2, //End of the current page
    "totalNum": -11, //total pages
    "pageNum": -1, //Current page number
    "dataList": [
      {
        "fromCount": 0,
```

```

        "guid": "20198002_20201212102810_20201212103810_1_mp4"
        "deviceNo": "20198002", //device id
        "channel": 1, //channel number, start from 1
        "createTime": "2020-11-01 12:22:11", //Task creation time
        "resSTime": "2020-11-01 13:22:11", //Recording start time
        "resETime": "2020-11-01 13:42:11", //Recording end time
        "resDeviceFilePath": "/mnt/123/345/3456/67/1221323.avi",
        "resWebPath": "/vssFiles/webdown/20198002/20201212102810_20201212103810_1.
        mp4",
        "downUrl": "http://172.16.50.210:9988/api/fileDown.php?filePath=RDovc29mdHdhcm
        UvaHRkb2NzL3Zzc0ZpbGVzL3dlYmRvd24vMjAxOTgwMDJfMjAyMTAzMTYwMDAwMDBfMjAyMTAz
        MTYwMDA4MDBfMS5tcDQ=&token=b791530c09905a8ef421e3a1c1b92521",
        "taskState": 1 //1-not finished, 2-finished
    }
}
]
}
}

```

downUrl composition:

1 filePath is a string (filePathBase64) after the base64 of the file path. The path supports full absolute path and web absolute path;

2 token, should be md5(privatekey+filePathBase64)

Examples of returned data in other cases

```

{
    "status": 10003, //status code, refer to Error code for HTTP interface
    "msg": "Server Error", //remind information
    "data": null
}

```

3.16 Passenger flow statistics report

3.16.1 Passenger flow details query

Interface address: <http://ip:9966/vss/passengerrecord/detail.action>

Request method: http Post

Request data format: json

Private request parameters:

Parameter	Type	length	Required?	Description	Sample value
pageNum	number		Yes	page number	2
pageCount	number		Yes	Number per page	20
devices	string		Yes	Device number, separated by multiple commas	300330,300331
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12

token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f
--------------	--------	--	-----	--------------	----------------------------------

Example of request json format:

```
{
  "pageNum": "2",
  "pageCount": "20",
  "token": "{{token}}",
  "devices": "99990001,99990002",
  "startTime": "",
  "endTime": ""
}
```

Private response parameters:

Returns the json array object files, the parameters of the array element object are as follows:

Parameter	type	length	Required?	Description	Sample value
deviceNo	string	64	Yes	device ID	
fleetName	string		Yes	Fleet name	
driverName	string		no	Driver's name	
createTime	string		no	Creation time	
reportTime	String		Yes	Reporting time	
gps	String		Yes	Location	
dw0	Integer		Yes	Number of people getting off at the front door	
Up0	Integer		Yes	Number of people getting on the front door	
Dw1	Integer		Yes	Number of people getting off the back door	
Up1	Integer		Yes	Number of people getting on the back door	
DwMid	Integer		Yes	Number of people getting off at the middle door	
upMid	Integer		Yes	Number of people getting on the middle door	
curTotal	Integer		Yes	Total number of people in the vehicle	
upTotal	Integer		Yes	Total number of people getting on	
dwTotal	Integer		Yes	Total number of people getting off	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.16.2 Passenger flow statistics query

Interface address:http://localhost:9966/vss/passengerrecord/statistics.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	type	length	Required?	Description	Sample value
pageNum	number		Yes	page number	2
pageCount	number		Yes	Number per page	20
devices	string		Yes	Device number, separated by multiple commas	300330,300331
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12
token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f

Example of request json format:

```
{
  "pageNum": "2",
  "pageCount": "20",
  "token": "{{token}}",
  "devices": "99990001,99990002",
  "startTime": "",
  "endTime": ""
}
```

Private response parameters:

Return the json array object list, the parameters of the array element object are as follows:

Parameter	type	length	Required?	Description	Sample value
deviceNo	string	64	Yes	device ID	
driverName	string		no	Driver's name	
dw0	Integer		Yes	Total number of people getting off at the front door	
Up0	Integer		Yes	Total number of people getting on at the front door	
Dw1	Integer		Yes	Total number of people getting off at the back door	

Up1	Integer		Yes	Total number of people getting on the back door	
upTotal	Integer		Yes	Total number of people getting on	
dwTotal	Integer		Yes	Total number of people getting off	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.17 Driver report

3.17.1 Query Driver's trip details

Interface address: http://localhost:9966/vss/driver/findDriverTripDetail.action

Request method: http Post

Request data format: json

Private request parameters:

Parameter	type	length	Required?	Description	Sample value
pageNum	number		Yes	page number	2
pageCount	number		Yes	Number per page	20
drivers	string		Yes	Driver card number, separated by multiple commas	test5555,test3333
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12
lang	string		Yes	language	zh_CN
token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f

Example of request json format:

```
{
  "pageNum": "2",
  "pageCount": "20",
  "lang": "zh_CN",
  "token": "{{token}}",
}
```

```

"drivers": "test5555,test3333",
"startTime": "",
"endTime": ""
}

```

Private response parameters:

Return the json array object list, the parameters of the array element object are as follows:

parameter	type	length	Required?	Description	Sample value
deviceID	string	64	Yes	device ID	
deviceName	string		Yes	Equipment name	
driverId	string		no	Driver id	
driverName	string		no	Driver's name	
startTime	String		Yes	Start time of the trip	
endTime	String		Yes	End time of the trip	
startGps	String		Yes	Longitude and latitude of the starting point, format: longitude, latitude	
endGps	String		Yes	Longitude and latitude of destination, format: longitude, latitude	
duration	String		Yes	Total travel time, format: HH:mm:ss	
mileage	String		Yes	Total mileage of trip	
speedAvg	String		Yes	Average speed	
speedMax	String		Yes	Maximum speed	
reportTime	String		Yes	Reporting time	
cardNo	String		Yes	card number	

Request returned data successfully

```

{
  "status": 10000,
  "msg": "Success",
  "data": {
    "list": [
      {
        "guid": "060D10CB-182D410D-A7E309A6-22573BA2",
        "deviceId": "999",
        "deviceName": "999",
        "driverId": "12232434",
        "driverName": null,
        "startTime": "2021-12-09 18:24:18",
        "endTime": "2021-12-09 18:28:50",
        "startGps": "22.558199,113.946419",
        "endGps": "113.946701,22.5581665",
        "duration": "00:04:32",
        "timer": null,
        "mileage": 0,

```

```

        "speedAvg": "0",
        "speedMax": "0",
        "reportTime": "2021-12-09 18:28:50.0",
        "cardNo": null,
        "totalAlarmN": 3,
        "totalAlarmDMS": 0,
        "totalAlarmADAS": 0
    },
    //....more
],
}
}

```

Examples of returned data in other cases

```

{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}

```

3.17.2 Driver's trip statistics query

interface address: http://localhost:9966/vss/driver/findDriverTripStatistics.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	type	length	Required?	Description	Sample value
pageNum	number		Yes	Page number, set to -1 when querying all	2
pageCount	number		Yes	Number per page, set to -1 when querying all	20
drivers	string		Yes	Driver card number, separated by multiple commas	test5555,test3333
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12
lang	string		Yes	language	zh_CN
token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f

Example of request json format:

```

{
  "pageNum": "2",
  "pageCount": "20",
  "lang": "zh_CN",
  "token": "{{token}}"
}

```

```

"drivers": "test5555,test3333",
"startTime": "",
"endTime": ""
}

```

Private response parameters:

Return the json array object list, the parameters of the array element object are as follows:

parameter	type	length	Required?	Description	Sample value
deviceID	string	64	Yes	device ID	
deviceName	string		Yes	Equipment name	
driverId	string		no	Driver id	
driverName	string		no	Driver's name	
quantity	String		Yes	Number of trips	
duration	number		Yes	Total travel time, in seconds	
mileage	number		Yes	Total mileage of trip	
maxMileage	number		Yes	Maximum mileage	
speedAvg	String		Yes	Average speed	
speedMax	String		Yes	Maximum speed	
reportTime	String		Yes	Reporting time	
cardNo	String		Yes	card number	

Request returned data successfully

```

{
  "status": 10000,
  "msg": "Success",
  "data": {
    "list": [
      {
        "deviceId": "999",
        "deviceName": "999",
        "driverId": "",
        "driverName": null,
        "quantity": 5,
        "mileage": 67,
        "duration": 1941,
        "maxMileage": 58,
        "speedAvg": "0.00",
        "speedMax": "5",
        "cardNo": null,
        "totalAlarmN": 55,
        "totalAlarmDMS": 0,
        "totalAlarmADAS": 0
      },
    ],
  }
}

```



```
}

```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.18 Geofence

3.18.1 Add Geofence group

Interface address: <http://localhost:9966/vss/area/addGroup.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
groupName	string		Yes	Group name	Test
lang	string		No	Language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f",
  "groupName": "test"
}
```

Response Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Group guid, (Unique ID)	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "guid": "16907E37-496043C0-969F4F0B-D29B15B4" //Return group guid
  }
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

}

3.18.2 Delete Geofence group

Interface address: http://localhost:9966/vss/area/deleteGroup.action

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
groupGuid	string		Yes	Group ID	16907E37-496043C0-969F4F0B-D29B15B4
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f",
  "groupGuid": "16907E37-496043C0-969F4F0B-D29B15B4"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.3 Modify Geofence Group

Interface address: http://localhost:9966/vss/area/updateGroup.action

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
groupGuid	string		Yes	Group ID	16907E37-496043C0-969F4F0B-D29B15B4
groupName	string		Yes	Name of new Group	NewTest
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f",
  "groupGuid": "16907E37-496043C0-969F4F0B-D29B15B4",
  "groupName": "NewTest"
}
```

```
"groupName": "NewTest"
```

```
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.4 Get details of Group

Interface address: <http://localhost:9966/vss/area/findGroup.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
groupName	String		No	The name of the searched group, if it is empty, it means to get all the groups	test
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "groupGuid": "F6D121D7-37B4406E-A012AA65-FA713461",
      "groupName": "Test"
    }
  ]
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

}

3.18.5 Add Geofence

Interface address: <http://localhost:9966/vss/area/insert.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaName	string		Yes	Name of Geofence	
areaType	number		Yes	Geofence Type, 1: polygon 2: route 3: circle	
description	string		No	Geofence Description	
areaInfo	string		Yes	Geofence location data (JSON)	{\"radius\":20848.064,\"gpsData\":[{\"longi\":113.88771,\"lati\":22.78631}]}
areaInfo.radius	number		No	The radius of the circle, if it is a polygon and a path, this field can not be set	
areaInfo.gpsData	array		Yes	latitude and longitude array	
groupGuid	string		No	ID of group	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "areaName": "test",
  "areaType": "1",
  "groupGuid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "description": "",
  "areaInfo":
  {\"radius\":0,\"gpsData\":[{\"longi\":113.9050568928826,\"lati\":22.60300073854558},{\"longi\":114.0162934651482,\"lati\":22.64124808536007},{\"longi\":114.02178662921068,\"lati\":22.55198755038766}]},
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b57"
}
```

Response Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Geofence guid, (Unique ID)	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
}
```

```

    "data":
    {
        "guid": "16907E37-496043C0-969F4F0B-D29B15B4"    //Return Geofence guid
    }
}

```

Example of return data in other cases:

```

{
    "status": 10003,
    "msg": "Server Error", //Notification info
    "data": null
}

```

3.18.6 Get all Geofence (including Group)

Interface address: <http://localhost:9966/vss/area/findAll.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
keyword	String		No	Search by Group name or Geofence name. Empty means get all	test
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```

{
    "token": "a559f5ff8af84f08b5e06075193b952f"
}

```

Response Parameter:

Returns a list of json array objects, and the parameters of the array element objects are as follows:

Parameter	Type	Length	Required?	Description	Example Value
groupGuid	string		Yes	Group ID	
groupName	string		Yes	Group name	
treeList	List		No	Geofence in group	
treeList.areaId	string		Yes	Geofence ID	
treeList.areaName	string		Yes	Geofence name	
treeList.areaType	number		Yes	Geofence Type, 1: Polygon 2: route 3: Circle	
treeList.description	string		No	Geofence Description	
treeList.createUsername	string		Yes	Creator	
treeList.creatoroleid	string		Yes	Role ID of creator	
treeList.groupName	string		No	Group name	
treeList.groupGuid	string		No	Group ID	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "groupGuid": "F6D121D7-37B4406E-A012AA65-FA713461",
      "groupName": "Test",
      "treeList": [
        {
          "areaId": "04660457-8B834435-A4FC2D52-EA251D1A",
          "areaName": "YandexTest",
          "areaType": 1,
          "description": "",
          "createUsername": "admin",
          "creatoroleid": "3c686465373948d08bc68bfe3b749b20",
          "groupName": "Test",
          "groupGuid": "F6D121D7-37B4406E-A012AA65-FA713461"
        }
      ]
    }
  ]
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.7 Modify Geofence

Interface address: <http://localhost:9966/vss/area/update.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Geofence ID	
areaName	string		Yes	Geofence name	
areaType	number		Yes	Geofence Type, 1: Polygon 2: route 3: Circle	
description	string		No	Geofence Description	
areaInfo	string		Yes	Geofence location data (JSON)	{"radius":20848.064,"gpsData":[{"longi":113.88771,"lati":22.78631}]}
groupGuid	string		No	Group ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "guid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "groupGuid": "5D05E347-3B1A-46C9-834E-33B43736B8D4",
  "areaType": "3",
  "areaName": "test2",
  "description": "",
  "areaInfo": "{\radius\":1516.5209044359387,\gpsData\":{ {\longi\":113.95087937229425,\lati\":22.561781465741564}}",
  "areaId": "E929E9C4-DAB14702-9604F5C4-88329C45",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.8 Delete Geofence

Interface address: <http://localhost:9966/vss/area/delete.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	String		Yes	Geofence ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "guid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.9 Get details of Geofence

Interface address: <http://localhost:9966/vss/area/find.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	String		Yes	Geofence ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "guid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}
```

Response Parameter:

Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Geofence ID	
areaName	string		Yes	Geofence name	
areaType	number		Yes	Geofence Type, 1: Polygon 2: route 3: Circle	
description	string		No	Geofence Description	
createUsername	string		Yes	Creator	
creatoroleid	string		Yes	ID of Creator's role	
groupName	string		No	Group name	
groupGuid	string		No	Group ID	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "guid": "3C6A7ABA-E48245B4-9B75456E-114233D8",
    "areaname": "GEO FENCE TEST 1",
    "areatype": 1,
    "areainfo": "{\\"radius\\":0,\\"gpsData\\":[{\\\"longi\\":113.14970634465196,\\"lati\\":22.902497017341833}]}",
    "description": "no enter",
    "createtime": "2022-07-14 15:20:01",
    "modifytime": "2022-07-14 15:20:01",
  }
}
```



```

    "creatoroleid": "FC628DCC-C16F43D4-B561BA47-4B1DE48C",
    "groupGuid": "37D0B1DE-7CAF4ABD-92CFA3A1-120D0CFF",
    "createUsername": "demo1"
  }
}

```

Example of return data in other cases:

```

{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}

```

3.18.10 Import Geofence from kml

Interface address: <http://localhost:9966/vss/area/import.action>

Request Method: http Post

Data format: multipart/form-data

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
file	File		Yes	file, supports kml or kmz	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

Return data after request succeeds:

```

{
  "status": 10000,
  "msg": "Success"
}

```

Example of return data in other cases:

```

{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}

```

3.18.11 Batch export to Geofence kml file

Interface address: <http://localhost:9966/vss/area/export.action>

Request Method: http post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guidList	string		Yes	Geofence ID, multiple IDs are separated by commas	E6AAD075-EAC94BE4-BB397971-A9978D43,CF8E279C-DE4947DD-AC0173CA-

					3199645F
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "guidList": "E6AAD075-EAC94BE4-BB397971-A9978D43,CF8E279C-DE4947DD-AC0173CA-3199645F",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}
```

Return data after request succeeds:

A kml file is returned, and the Content-Type of the response body is: application/vnd.google-earth.kml+xml

The transfer uses block encoding: Transfer-Encoding: chunked

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.12 Issue Geofence to devices

Interface address: <http://localhost:9966/vss/areaDevice/insert.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string		Yes	Geofence ID	
deviceId	string		Yes	Device ID	
toggleConditionJson	string		Yes	Send configuration Parameter (JSON). For configuration Parameter, please refer to the H-protocol document	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "areaId": "3C6A7ABA-E48245B4-9B75456E-114233D8",
  "deviceId": "028073",
  "toggleConditionJson": {
    "over": {
      "enable": 0,
      "spd": "",
      "dut": "",
      "linkRecord": 0,
      "linkLockChn": 0,
      "linkUploadChn": 0,

```

```

    "linkOutput": 0,
    "linkSnapChn": 0,
    "linkNet": 0,
    "linkCamSwitch": 0,
    "linkTTSSwitch": 0,
    "linkBuzzer": 0
  },
  "st1": "00:00:00",
  "et1": "23:59:59",
  "st2": "",
  "et2": ""
},
"token": "4cb2e1c693254eff87e038db3662213a"
}

```

Response Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Guid of Issue record, unique ID of the mapping relationship between Geofence and device	

Return data after request succeeds:

```

{
  "status": 10000,
  "msg": "Success",
  "data":
  {
    "guid": "16907E37-496043C0-969F4F0B-D29B15B4"    // Guid of Issue record
  }
}

```

Example of return data in other cases:

```

{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}

```

3.18.13 Get issue record of a geofence

This interface is used to obtain which devices a Geofence is issued to, and the detailed parameters issued to each device;

Interface address: /vss/areaDevice/getAreaListByAreaId.action

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string		Yes	Geofence ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "areaId": "3C6A7ABA-E48245B4-9B75456E-114233D8",
  "token": "4cb2e1c693254eff87e038db3662213a"
}
```

Response Parameter:

Returns a list of json array objects, and the parameters of the array element objects are as follows:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string			Geofence ID	
guid	string			Guid of Issue record, unique ID of the mapping relationship between Geofence and device	
deviceid	string			Device ID	
createtime	string			Issue time	
modifytime	string			Modify 时间	
devicename	string			Device name	
areaSendState	number			Status: 0: not synched 1: synched	
areaSendErrCode	string			error code, 0: not issued or issued successfully 1: issuing timed out -1: json generation failed 99: other errors	
toggleconditionjson	string			configuration Parameter of issuing (JSON). For configuration Parameter, please refer to the H-protocol document	
lastToggleConditionJson	string			Last configuration Parameter of issuing (JSON). For configuration Parameter, please refer to the H-protocol document	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "guid": "74A5CD7F-C779460D-8ABF3CB0-85BE2DD2",
      "areaId": "3C6A7ABA-E48245B4-9B75456E-114233D8",
      "deviceid": "028073",
      "toggleconditionjson":
      "{\n\"over\":{\n\"enable\":0,\n\"spd\":\"\n\", \n\"dut\":\"\n\", \n\"linkRecord\":0,\n\"linkLockChn\":0,\n\"linkUploadChn\":0,\n\"linkOutput\":0,\n\"linkSnapChn\":0,\n\"linkNet\":0,\n\"linkCamSwitch\":0,\n\"linkTTSSwitch\":0,\n\"linkBuzzer\":0}\n\",
      "createtime": "2022-07-14 17:02:39",
      "modifytime": "2022-07-14 17:02:39",
      "devicename": "028073",
      "areaSendState": 1,
      "areaSendErrCode": 0,
    }
  ]
}
```

```

    "lastToggleConditionJson":
    "{\over\":{\enable\":0,\"spd\":\"\",\"dut\":\"\",\"linkRecord\":0,\"linkLockChn\":0,\"linkUploadChn\":0,\"linkOutput\":0,\"linkSnapC
hn\":0,\"linkNet\":0,\"linkCamSwitch\":0,\"linkTTSSwitch\":0,\"linkBuzzer\":0}"
  }
}

```

Example of return data in other cases:

```

{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}

```

3.18.14 Delete a Geofence in device

Interface address: <http://localhost:9966/vss/areaDevice/delete.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	ID of issue record. Note: This is not the Geofence id, but the guid returned by the Geofence interface	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```

{
  "guid": "3C6A7ABA-E48245B4-9B75456E-114233D8",
  "token": "4cb2e1c693254eff87e038db3662213a"
}

```

Return data after request succeeds:

```

{
  "status": 10000,
  "msg": "Success"
}

```

Example of return data in other cases:

```

{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}

```

3.18.15 Get the issue details of a Geofence

This interface is used to obtain the detailed parameters of issuing a Geofence to a device.

Interface address: <http://localhost:9966/vss/areaDevice/getlastToggleConditionJson.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string		Yes	Geofence ID	
deviceId	string		Yes	Device ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "areaId": "3C6A7ABA-E48245B4-9B75456E-114233D8",
  "deviceId": "20160604",
  "token": "4cb2e1c693254eff87e038db3662213a"
}
```

For the specific definition of the returned data, please refer to the H-protocol document.

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "over": {
      "linkSnapChn": 0,
      "dut": "",
      "linkOutput": 0,
      "linkCamSwitch": 0,
      "enable": "1",
      "spd": "",
      "linkUploadChn": 1057,
      "linkBuzzer": 0,
      "linkTTSSwitch": 0,
      "linkLockChn": 1057,
      "linkNet": 0,
      "linkRecord": 0
    }
  }
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

4 Websocket real time information Interface

Before starting WebSocket subscription for real-time business messages, it must also log in through the HTTP login interface, which requires to obtain the PID parameter and token parameter returned by the server from the login interface and use them for WebSocket subscription requests.

Subscribe to real-time information including device online or offline, alerts/events, GPS status, and more.

4.1 ws login

Interface Address: ws://192.168.1.2:36300/ws

Request way: websocket, short as 'ws'

Request direction: ws client->ws server

Private request parameter:

Json Object/Attribute	Type	Length	Required	Description	Example value
action	number		Yes	ws message type, Message type, login and subscription type is 80,000, see: ws message type table	80000
Payload object:					
payload.username	string	64	Yes	User name	admin
payload.pid	string	128	Yes		b9c7f86sdeg6f7e4799cefgaaeed465dc1cc704cb1db207bdab9c1758ad39bd187d350fa949b2ff9ddda8436467b1
payload.token	string	64	Yes	The token obtained after a successful API login	0db6acgf92d0a2bds5702d92ef906631

Json format example:

```
{
  "action": "80000",
  "payload": {
    "username": "admin",
    "pid": "b9c7f86f5a21dc6f7e4799cde801eed465dc1cc704cb1db207bdab9c1758ad39bd187d350fa949b2ff9ddda8436467b1",
    "token": "0db6d4b92d0a2b0c45702d92ef906631"
  }
}
```

Request successfully and return data

```
{
  "action": "80000",
  "payload": {
    "result": "success",
    "msg": "success"
  }
}
```

```
}

```

Return data example in the other situation

```
{
  "action": "80000",
  "payload": {
    "result": "fail",
    "msg": "Invalid username"
  }
}
```

4.2 ws subscribe to real-time message

interface address: ws://192.168.1.2:36300/ws

Request method: websocket, also named as ws

Request direction: ws client->ws server

Private request parameters:

Json Object	Type	Len	Require	Description	Example
action	number		Yes	ws message type, subscription type is 80001, refer: ws message type table	80001
Payload Object: Empty					

Json format example:

```
{
  "action": "80001",
  "payload": ""
}
```

4.3 Heartbreak message

Interface address: ws://192.168.1.2:36300/ws

Request direction: ws client->ws server

Client should send the heartbreak message to server per 1 minute to make sure channel active.

Private request parameter:

Json Object/Attribute	Type	Length	Require	Description	Example Value
action	number		Yes	ws message type, login and subscription type is 80009, see: ws message type table	
Payload Object:					
payload.username	string	64	Yes	User name	admin
payload.token	string	64	Yes	The token obtained after a successful API login	0db6acgf92d0a2bds5702d92ef906631

Json format example:


```
{
  "action": "80009",
  "payload": {
    "username": "admin",
    "token": "0db6d4b92d0a2b0c45702d92ef906631"
  }
}
```

Request successfully and return data

```
{
  "action": "80009",
  "payload": {
    "result": "success",
    "msg": ""
  }
}
```

4.4 Push device status information

Request way : websocket

Request direction: ws server->ws client

Request parameter:

Json Object/Attribute	Type	length	Required	Description	Example value
action	number		Yes	ws message type, push status message Yes80003, see: WS message type table	80000
Payload object definition					
payload.deviceID attribute	string	128	Yes	Device id	99990001
payload.location positioning information object					
location.mode	Number	8	Yes	Location type 0—invalid positioning 1--GPS 2--BD 3---Glonass 4--AGPS 5—base station location 6--WIFI	1
location.dtu	String	24	Yes	GPS positioning time	2018-01-11 11:35:02
location.direct	Number	8	Yes	Vehicle direction, 0~359°	
location.speed	Number		Yes	Speed, unit: KM/H	
location.altitude	String		Yes	Altitude(meter)	

location.precision	Number		Yes	Positioning precision(meter)	
location.longitude	String		Yes	longitude	114.106476
location.latitude	String		Yes	latitude	22.5767326
payload.module Module state object					
module.mobile	Number			Mobile Network module status, (0:the unknown,1:work,2:abnormal,3:not exist)	
module.location	Number			Positioning module status, (0:Unknown,1:Work, 2:Strange,3:Not exist)	
module.record	String			Channel recording video status hexadecimal string with each bit representing the recording state of the channel, starting at 0.For example: Bit0 corresponds to the first video, and so on	0xf
payload.basic Object					
basic.key	Number			1 acc on; 0 acc off	
payload.storage Stores an array of state objects					
storage.index	Number			Disk serial number	
storage.status	Number			Disk status,0:Unknown, 1:Recording, 2:Free, 3:Strange, 4:Disk is full	
storage.total	Number			Total storage(MB)	
storage.free	Number			The rest of storage(MB)	
payload.mileage Mileage object					
mileage.todayDay	Number			On the day of the mileage	
mileage.total	Number			Total mileage	

Json Format example:

```
{
  "action": "80003",
  "payload": {
    "deviceId": "300341", //Device name
    "location": { //Positioning information
      "mode": "1", //Positioning type, 1 is GPS
      "dtu": "2018-01-11 11:35:02", //Device time
      "direct": "59", //Direction 0~359°
      "satellites": "0", //Satellite number
    }
  }
}
```

```

        "speed": "17", //Speed, unit: KM/H
        "altitude": "21", //Altitude(meter)
        "precision": "50", //Positioning precision(meter)
        "longitude": "114.106476", //Longitude
        "latitude": "22.5767326" //Latitude
    },
    "module": { //State of the module
        "mobile": "1",
        "location": "1",
        "wifi": "1",
        "gsensor": "1",
        "record": "0xf" //Channel recording state, BIT0 corresponds to the first video, and so
on
    },
    "basic": {
        "key": "1", //1 acc on; 0 acc off
    }
    "storage" : [ //Stores an array of state objects
    {
        "index" : "0", //Serial number
        "status" : "1", //Disk status
        "total" : "60906", //Total storage(MB)
        "free" : "0" //the rest of the storage(MB)
    }
    ],
    "mileage": { //mileage information
        "todayDay": "0", //On the day of the mileage(meter)
        "total": "0" //Total mileage(meter)
    }
}
}

```

4.5 Push Device online/offline info

Request way: websocket

Request direction: ws server->ws client

Request parameter:

Json object/attribute	Type	length	Required	Description	Example value
action	number		Yes	ws message type, online/offline message Yes80005, see: ws message type table	80000
Payload object definition					
payload.deviceID attribute	string	128	Yes	Device id	99990001
payload.accessMode	Number		Yes	Device onlinenetwork type, see: device network status table	

Json example:

```
{
  "action": "80005",
  "payload": {
    "deviceID": "300342", //Device name
    "accessMode": 5 Device online network type
  }
}
```

4.6 Device event message push (Alarm message)

Request way: websocket

Request direction: ws server->ws client

Request parameter:

Json object/attribute	Type	length	Required	Description	Example value
action	number		Yes	ws message type, event message push Yes80004, see ws message type table	80004
Payload object definition					
payload.deviceID attribute	String	64	Yes	Device id	99990001
payload.alarmID attribute	string	48	Yes	Event uuid	60D27AA2-E2E7495A-B77AB878-D58861F2
payload.payload event details object					
payload.dtu	String	24	Yes	Occurrence device time	2016-01-11 11:35:02
ec	Number	16	Yes	Type, see event code table	
St	String	24	Yes	start time	2016-01-11 11:35:02
et	String	24	Yes	end time	2016-01-11 11:35:12
det	Object		No	Alarm detailed parameters. Contains detailed alarm related parameters and alarm subtypes. You can refer to the code table.	{"num":"sd1","st":"0"}。
payload.location Object (location message object)					
location.mode	Number	8	Yes	Location type 0—invalid 1--GPS 5--BD 6---Glonass 7--AGPS	1

				5—base station 6—WIFI	
location.dtu	String	24	Yes	GPS locating time	2018-01-11 11:35:02
location.direct	Number	8	Yes	direction, 0~359°	
location.speed	Number		Yes	speed, unit: KM/H	
location.altitude	String		Yes	Altitude (meter)	
location.precision	Number		Yes	Location accuracy (meter)	
location.longitude	String		Yes	Longitude	114.106476
location.latitude	String		Yes	Latitude	22.5767326

Json example:

```
{
  "action": "80004",
  "payload": {
    "deviceID": "300342", //device id
    "alarmID": "60D27AA2-E2E7495A-B77AB878-D58861F2", //event id
    "payload": {
      "det": { //detail
        "ch": "1" //Triggered channel, starting from 1
      },
      "dtu": "2018-01-12 09:34:20", //Device time
      "ec": "1", //Event message type
      "et": "2018-01-12 09:34:30", //end time
      "st": "2018-01-12 09:34:20"
    }
  },
  "location": {
    "mode": "1",
    "dtu": "2021-01-25 17:31:59",
    "direct": "0",
    "satellites": "0",
    "speed": "19",
    "altitude": "1833",
    "precision": "5",
    "longitude": "114.111992",
    "latitude": "22.5380726"
  },
}
```

5 Web video integration (iframe embedding method)

Iframe web page embedding supports online playback of H.264 and H.265

5.1 Device real-time video preview URL

External platforms can load the VIDEO preview HTTP URL provided by VSS through iframe in their own Web pages, and pass relevant parameters through URL to quickly complete the development of video preview function. This page supports multiple channels playing at the same time.

- ✓ Real time preview http URL address

Format example is as follows:

http://vssServerIp:9966/vss/apiPage/RealVideo.html?token=adfsdfabdddabcdefg&deviceId=20191122&chs=1_3&stream=0&wnum=4&panel=1&buffer=2000

The parameters are defined as follows:

- (1) token, required parameter, Yes the string of tokens returned after successful login through the HTTP API login interface
- (2) deviceId, required parameter, the device ID which needed to play video
- (3) chs, required parameter, channel list, each channel number is separated by an underscore” _ ”
- (4) stream, required parameter, Stream code, 1 represents the main-stream, 0 represents the first sub-stream
- (5) wnum, optional parameter, Default window number (Required is one of 1/4/6/9/16), when less than the number of channels, the number of channels shall prevail.
- (6) panel, optional parameter, whether to display the window layout panel. 1.display(default), 0 not display.
Note: if needs to control multiple screens by yourself, you can open one channel at a time, and make a configuration wnum=1,panel=0.
- (7) buffer, optional parameter, video buffer time, in milliseconds, default 2000. For mobile app, it is recommended to set to 0
Note: If needs to control multiple channels, you can open one channel at a time and set wnum=1,panel=0

- ✓ Real time preview https URL address

All of them are the same except that the port is different (using the default port of 443),as follows:

https://vssServerIp/vss/apiPage/RealVideo.html?token=adfsdfabdddabcdefg&deviceId=20191122&chs=1_3&stream=0

5.2 Web URL for device to listen

- ✓ Device to monitor the http URL address

The format example is as follows:

<http://vssServerIp:9966/vss/apiPage/RealListen.html?token=abcdefg&deviceId=20191122&chs=1>

The parameters are defined as follows:

- (1) Token parameter, the string of tokens returned after successful login through the HTTP API login interface
- (2) deviceId parameter, the device ID of the device which needs to play
- (3) chs parameter, channel number, starting at 1, the listener can carry only one channel number

- ✓ https URL address

- ✓ All of them are the same except that the port is different (using the default port of 443),as follows:

5.3 URL of device intercom webpage

Web intercom can only be used normally under https. VSS is a self-signed certificate by default, and the browser will prompt an access warning. It is recommended to install a certificate issued by a certificate authority.

- ✓ Device intercom https URL address

- ✓ The format example is as follows:

<https://vssServerIp:36301/vss/apiPage/dialog.html?token=abcdefg&deviceId=20191122>

The parameters are defined as follows:

- (1) The token parameter is the token string returned after successful login through the http api login interface
- (2) deviceId parameter , the device ID of the device tht needs to play

5.4 Web URL for Device playback video

This page supports simultaneous playback of multiple videos

- ✓ Device video replay HTTP URL address

The format example is as follows:

http://vssServerIp:9966/vss/apiPage/ReplayVideo.html?token=abcdefg&deviceId=20191122&chs=1_3&st=20200810110900&et=20200810112800&wnum=4&panel=1&speed=4&buffer=3000

The parameters are defined as follows:

- (4) token, required parameter , The string of tokens returned after successful login through the HTTP API login interface
- (5) deviceId, required parameter , the device ID of the device which needs to play
- (6) chs, required parameter , A list of channels, with each channel number separated by an underscore "_"
- (7) st, required parameter , Video start time. Format YYYYMMDDHHIISS
- (8) et, required parameter , Video end time
- (8) wnum, optional parameter , Default window number (Required is one of 1/4/6/9/16), when less than the number of channels, the number of channels shall prevail.
- (9) panel, optional parameter , Whether to display the window layout panel.1, displayed (default), 0 not displayed
- (10) Speed, fast-forwarding parameter, could be 2/4/8/16
- (11) buffer, optional parameter, video buffer time, in milliseconds, default 2000. For mobile app, it is recommended to set to 0

- ✓ https URL Addresss

All of them are the same except that the port is different (using the default port of 443),as follows:

5.5 Web URL of playback of small alarm video and Server Recording clips

The small alarm video in server refers to the small video clip generated by the alarm linkage, which is finally stored on the VSS server. This webpage can only play one server video file, and does not support multiple channel play at the same time.

Server video clips: The video server actively grabs the real-time video of the device and saves it on the VSS server. The video is sliced and stored by time. This interface supports playing a certain clips of video. The video path can be obtained using the http video file search interface.

- ✓ Server video playback http URL address

Example:

<http://vssServerIp:9966/vss/apiPage/ReplayAlarmServerVideo.html?token=abcdefg&deviceId=20191122&chs=1>

&fpath=RSUzQSU1Q0hvd2VuJTVDRG93bmxxvYWQlNUMyMDIwLTA4LExJTVDMjAxOTgwMDIINUNjaDAxXzIwMjAwODExXzAwMzAwMF8wMDUzMDJfMDEuYXNm

The parameters are defined as follows:

- (1) The token parameter is the token string returned after successful login through the http api login interface
- (2) deviceId parameter, the device id of the device to be played
- (3) chs parameter, channel list, each channel number is separated by underscore "_"
- (4) The fpath parameter, the full path of the server video file obtained through the http video search interface, needs to be passed in after base64 encoding

- ✓ https URL address

Except that the port is different (using the default port 443), everything else is the same.

5.6 Web URL for Server to playback of regularly downloaded video

Server video refers to the video downloaded from the device by VSS on a regular basis. This page can only play a server video file, and does not support multiple images.

- ✓ Video replay HTTP server URL

The format example is as follows:

<http://vssServerIp:9966/vss/apiPage/ReplayServerVideo.html?token=abcdefg&deviceId=20191122&chs=1&fpath=RSUzQSU1Q0hvd2VuJTVDRG93bmxxvYWQlNUMyMDIwLTA4LExJTVDMjAxOTgwMDIINUNjaDAxXzIwMjAwODExXzAwMzAwMF8wMDUzMDJfMDEuYXNm>

The parameters are defined as follows :

- (1) Token parameter, the string of tokens returned after successful login through the HTTP API login interface
- (2) deviceId parameter, the device ID of the device which needs to play
- (3) chs parameter, A list of channels, with each channel number separated by an underscore "_"
- (4) fpath parameter, the full path of the server video file obtained through the HTTP video search interface, need to be encoded by base64 and then write.

- ✓ https URL address

All of them are the same except that the port is different (using the default port of 443)

6 Web video SDK (Javascript SDK)

Third-party platforms can realize online playback of various audio and video data by integrating VSS's open JavaScript player SDK. **Support both H.264, and H.265**

SDK api document address: http://vssServerIp:9966/vss/apiPage/api_vss.html

SDK Demo address: http://vssServerIp:9966/vss/apiPage/api_demo_player.html

7 Video Request HTTP-FLV URL

The Vss platform provides streaming media request services based on the flv over http standard protocol for device video access, and the audio is encoded by AAC. Third-party platforms can request playback through common open source players such as flvjs and vlc.

Note: This solution is not recommended. This solution has a general experience, poor compatibility, and does not support H265.

1. Device real-time video request url

http://vssServerIp:33122/live?token_deviceId_channel_stream

Service port 33122, service type flag is "live"

The url needs to carry the following parameters, and each parameter is separated by underscore:

- token, Token obtained after logging in through the http web api login interface
- deviceId, Device ID
- channel, Channel number, starting from 1.
- stream, The requested stream type, 1 represents the main stream, 0 represents the sub stream

Example:

Request the first channel sub-stream audio and video stream with the device id of 666888:

http://172.16.50.210:33122/live?a1512bf437d6c0a086d9881ba277d7fc_666888_1_0

2. Device video playback request url

http://vssServerIp:33122/replay?token_deviceId_channel_startDateTime_endDateTime

Server port 33122, service type flag is "replay"

Url needs to carry the following parameters, and each parameter is separated by underscore:

- token, Token obtained after logging in through the http web api login interface
- deviceId, Device ID
- channel, Channel number, starting from 1.
- startDateTime, Request recording start time, format: yyyyMMddHHmmss, for example 20201015123040, which means 2020-10-15 12:30:40
- endDateTime, Request recording end time, same format as above

Example:

http://172.16.50.210:33122/replay?a1512bf437d6c0a086d9881ba277d7fc_666888_1_20200810110900_20200810112800

8 Code sheet

8.1 Error code for HTTP interface

Error code	Description
10000	Operation successful
10001	Account or password error
10002	Verification code error
10003	Server error
10004	Not logged in
10005	Please fill in the mandatory parameter
10006	Device not exist
10007	Operation failed
10008	Vehicle not exist
10009	Fleet not exist
10010	Driver not exist
10011	Role not exist
10012	Account not exist
10013	Server not exist
10014	Area not exist
10015	Vehicle not in the area
10016	Account already existed
10017	Account not activated
10018	No authority
10019	Auto download plan not exist
10020	Alarm auto download plan not exist
10021	File Information of alarm auto download file not exist
10022	File information of auto download not exist
10023	Access token error
10024	Record not exist
10025	No record
10026	Illegal parameter
10027	Original password error
10028	Fail to update configuration of server
10029	Device already exist
10030	Quantity of intercom channel cannot be more than 32.
10031	Quantity of output channel cannot be more than 32.
10032	Quantity of input channel cannot be more than 32.
10033	Quantity of audio coding channel cannot be more than 32.
10034	The number of audio coding channel does not match the number of channel name.
10043	Device is offline

10036	rule does not exist
10037	The maximum device ID cannot exceed 9999
10038	license has expired
10040	Automatic download task does not exist
10041	Device IDs are inconsistent
10042	Device type does not exist
10047	unknown error
10048	Exceeds the number of licenses
10049	The file format is wrong, only png format is supported
10050	area already in use

8.2 Device online status

Value	Description
-1	Offline
0	Unknown
1	Ethernet
2	WIFI
3	2G
4	3G
5	4G
6	5G
7	WIFI+3/4/5G, connected to mobile network agent via Wi-Fi
8	CABLE+3/4/5G, connected to mobile network agent via Ethernet

8.3 Type of data frame

Value	Description
0	Invalid
1	Main Stream (H264)
2	Virtual stream (H264)
3	Audio frame (G726 contains 2 bytes special header)
4	Serial port data frame
5	File data frame(Media data length and other information are all 0, indicating that the file transmission is complete and the video playback is complete)
6	Status data frame(Valid when played back)
7	Alarm data frame(Valid when played back)

8.4 Type of service

Value	Description
0	Invalid
1	Management
2	Gateway

3	Storage
4	Flow media

8.5 Device status

Value	Description
-1	Offline
0	Unknown
1	Online via Ethernet
2	Online via WIFI
3	Online via 2G mobile network
4	Online via 3G mobile network
5	Online via 4G mobile network
6	Online via 5G mobile network
7	Online via WIFI+3/4/5G, connected to mobile network agent via Wi-Fi
8	Online via CABLE+3/4/5G, connected to mobile network agent via CABLE

8.6 Type of file

Value	Description
0	Unknown
1	Ordinary video
2	Alarm video
3	Ordinary captured file
4	Alarm captured file
5	Update file
6	Log file
7	Configuration file
8	Black box

8.7 PTZ movement code

Value	Description
0	Unknown
1	Up
2	Down
3	Left
4	Right
5	Upper left
6	Lower left
7	Upper right
8	Lower right
9	Jump to preset position

10	Set preset position
11	Clear preset position
12	Big aperture
13	Small aperture
14	Zoom out
15	Zoom in
16	Focus near
17	Focus far
18	Auto pan
19	Start wiper
20	Stop wiper
21	Cruise on
22	Cruise off
23	Light on
24	Light off

8.8 Type of device

Value	Description
Hero-ME41-04	Howen device, Hero-ME41-04
Hero-ME40-04	Howen device, Hero-ME40-04
Hero-ME34-04	Howen device, Hero-ME34-04
Hero-ME32-04	Howen device, Hero-ME32-04
Hero-ME31-04	Howen device, Hero-ME31-04
Hero-ME34-08	Howen device, Hero-ME34-08
Hero-ME31-08	Howen device, Hero-ME31-08
Hero-MA80-08	Howen device, Hero-MA80-08
Hero-ME41-02	Howen device, Hero-ME41-02

8.9 Event code

Value	Description
0	Unknown
1	Video lost
2	Motion detection
3	Video overshadowed
4	Alarm input triggered
5	Alarm of emergency
6	Alarm of low speed
7	Alarm of high speed
8	Alarm of low temperature
9	Alarm of high temperature
10	Alarm of humidity

11	Park overtime
12	Alarm of vibration
13	Electronic fence
14	Electronic route
15	Exception of open or close door
16	Storage abnormal
17	Fatigue driving
18	Exceptional volume of gas
19	Illegal ignition
20	Location module abnormal
21	Front panel open
22	RFID tagged
23	IBUTTON
24	Rapid acceleration
25	Rapid deceleration
26	Low speed pre-alarm
27	High speed pre-alarm
28	Voltage alarm
29	Population statistics
30	DMS/ADAS alarm (Active safety alarm)
31	Acc on
32	idle speed
33	Gps antenna disconnected
34	Gps antenna short circuit
35	IO output
36	CANBus connection abnormal
37	Towing alarm
38	Glide in neutral
39	RPM too high
40	vehicle move
41	Trip start (st/et/dtu time is the same)
42	In trip
43	End of the trip (reported periodically after acc off)
44	GPS location recovery
45	Abnormal video
768	End of trip notification
769	Upgrade notification
101	Forward collision warning
102	Lane departure warning
103	Front vehicle distance is too close warning
104	Pedestrian collision warning

105	Frequent lane change alarm
106	Road marking violation alarm
107	Harsh acceleration
108	Harsh breaking
109	Road sign recognition incident
110	FCW Forward relative velocity collision
111	HMW Forward absolute velocity collision
112	LDW_L Left lane departure
113	LDW_R Right lane departure
114	VB Low-speed forward collision alarm
115	Fatigue driving alarm
116	Calling alarm
117	Smoking alarm
118	Distracted driving alarm
119	Driver abnormal alarm
120	Driver change event
121	Eyes closing
122	Yawning
123	Camera cover alarm
124	glance right and left
125	Not wearing a seat belt
126	No driver
127	Drinking water
128	Driver changing
129	Driver returns
130	Infrared sunglasses
131	Driver authentication succeeded
132	Driver authentication failed
133	No face detected
200	enter the fence
201	leave the fence
202	Overspeed alarm in fence
203	Overspeed warning inside the fence
204	Low speed alarm in fence
205	Low speed warning inside the fence
207	Engine on in fence
208	Engine off in fence
209	Timeout inside the fence
210	Refuel
211	Fuel Theft
212	Rapid acceleration (gsebsor)

213	Rapid deceleration (gsebsor)
214	sharp turn (gsebsor)
215	entry fence warning
216	Leave the fence warning
10001	Insufficient disk space on server
11001	Excessive device traffic
11003	The device is offline for a long time

8.10 DMS/ADAS alarm subtype

The definition of the subtype is encapsulated in the alarm details *det* object in the DMS/ADAS alarm message.

Example of det object format:

```
{
  "tp": "70"
}
```

tp: subtype, descriptions as below:

Remarks: 1 to 21 are ADAS types; 33 to 83 are DMS types;

Value	Description
1	Forward collision alarm
2	Lane departure warning
3	Car distance too near
4	Pedestrian collision alarm
5	Frequent lane change alarm
6	Road sign violation alarm
7	Harsh acceleration
8	Harsh braking
16	Road sign recognition event
17	FCW: forward collision warning
18	HMW: Headway monitoring warning
19	LDW_L: left lane departure warning
20	LDW_R: right lane departure warning
21	VB: low-speed forward collision warning
33	Fatigue driving alarm
34	Phone call alarm
35	Smoking alarm
36	Distracted driving alarm
37	Driver abnormal alarm
49	Driver Changed
65	Eye closed
66	Yawning

67	Camera cover
68	Distracted Driving
69	Seat belt not fastened
70	No driver
71	Drinking water
72	Driver shift
73	Driver back
80	Infrared sunglasses
81	Driver ID identified successfully
82	Driver ID identified failed
83	Cannot detect human face

8.11 Alarm details type

video loss, motion detection, video blind, input trigger, emergency alarm		
content	filed name	description
channel	ch	trigger channel, starting from 1

Input trigger		
Content	Field	Description
Channel	ch	Triggered channel, starting from 1
Number	num	0- MDVR box open. 1- Emergency/ Panic 2- F-door 3- M- door 4- B-door 5- Near light 6- Far light 9- R-Turn (right turn) 10- L-Turn (left turn) 11- Braking, 12- Reverse 13- Reserved 1 14- F-door close 15- M-Door Close 16- B-door close 17- Talk (start the intercom) 18- Raise up 19- Airtight 20- Load 31-IBT2

low speed alarm, over speed alarm, low speed warning, high speed warning, harsh acceleration, harsh braking, low temperature alarm, high temperature alarm

content	filed name	description
trigger threshold	vt	conditional value, if >90 is over speed, then the trigger threshold is 90
time threshold	tt	if > 90 over 5 second is over speed, then the time threshold is 5 second
maximum value	max	Maximum value during the alarm
minimum value	min	Minimum value during the alarm
average	avg	Average value during the alarm
current value	cur	Current value when reporting
Previous second	Pre	The value of one second before

Overtime parking		
content	filed name	Description
trigger value	vt	
parking time	st	second

vibration alarm / Acceleration Alarm		
content	field name	Description
trigger threshold	vt	conditional value, if >90 is over speed, then the trigger threshold is 90
time threshold	tt	if > 90 over 5 second is over speed, then the time threshold is 5 second
maximum value	max	Maximum speed value during the alarm (speed*100)
minimum value	min	Minimum speed value in a period (speed*100)
current value	avg	Average speed value in a period (speed*100)
current value	cur	Current speed value when reporting
direction	dt	1—X direction, 2—Y direction, 3—Z direction, 4—impact, 5—tilt, 6-turn, 7-Harsh acceleration, 8—harsh braking
Previous second	pre	The value of one second before

Electronic fencing, Electronic route		
content	field name	description
numbering	num	fencing or route numbering
status	st	0—enter, 1—leave 2- fence over-speed
Keep field	Resv	At present, resV has value only when St is equal to 2, 1 means the beginning of overspeed, 0 means the end of overspeed

abnormal open/ close door		
content	field name	description

Channel	Ch	Triggered channel of inputs, starting from 1
numbering	num	2—front door, 3—mid door, 4—back door
status	st	0—close, 1—open

storage abnormal		
content	filed name	description
numbering	num	For example: sd1, sd2, hdd1, hdd2
status	st	0—loss, 1—broken, 2— cannot overwrite, 3—Write block fail, 4—disk broken

People counting		
content	filed name	description
Front door onbus people number	Up0	The number of onbus people, <0 means invalid
Front door offbus people number	Dw0	The number of offbus people, <0 means invalid
Back door onbus people number	Up1	The number of onbus people, <0 means invalid
back door offbus people number	Dw1	The number of offbus people, <0 means invalid
longitude	lon	
latitude	lat	
	Pat	0-real time 1-stored and forward (data was resent after network works)
time	tm	Format: “ yyyy-mm-dd hh:mm:ss”
	Va	Current onbus people number byte is valid or not (device calculated), 0-invalid , 1-valid
	Cur	the current number of people onbus

Active Safety (DMS & ADAS)		
content	field name	description
fatigue level	de	Refer to alarm type

fuel level abnormal alarm		
content	field name	description
trigger threshold	vt	
oil tank capacity	to	unit : Litre

balance fuel capacity	fr	unit : Litre

Swipe Card (RFID/NFC/ Magnetic Card reader etc)		
Content	Field name	Description
Swipe card type	tp	1-Driver, 2-Student, 3-invalid card
Swipe card info	cn	Card number, etc.
Onboard/offboard	up	1— Onboard (check-in), 2—Offboard (check out), else—invalid if for Driver swipe card: 1-login, 2-log out. Other-invalid
History	ht	1—Historical data 2—Realtime data, Else—invalid

Voltage		
Content	Field name	Description
High/low voltage	dt	1—low voltage 2—high voltage

Fatigue detection (built-in DMS) (Only for firmware with built-in fatigue function)		
Content	Field name	Description
Fatigue detection	appDefinedCode	1133Fatigue alarm (close eye, shake head, Overlooking the mobile phone and etc)

8.12 ws Message type table

Value	Description
80000	The client will request login verification after successful connection, and automatically start subscription after successful connection
80001	Subscribe to real-time messages
80003	The server pushes device status information to the client
80004	The server pushes device alarm event to the client
80005	The server pushes device online/offline information to the client
80009	The client sends heartbeat information to server at regular intervals